

Week 1

- **Problem 1: Conjunction introduction.**

Let $a, b \in \mathbb{R}$. Suppose that

$$a + b = 8, \quad ab = 12.$$

Prove that

$$a > 0, \quad b > 0 \quad \text{and} \quad |a - b| = 4.$$

- **Solution 1.1**

Since a and b satisfy $a + b = 8$ and $ab = 12$, they are the roots of the quadratic equation:

$$t^2 - 8t + 12 = 0$$

Factoring the equation gives:

$$(t - 2)(t - 6) = 0$$

Thus, the roots are 2 and 6. This implies that either $\{a, b\} = \{2, 6\}$.

In both cases, we can check each part of the conjunction:

1. $a > 0$ and $b > 0$ since both $2 > 0$ and $6 > 0$ are true.
2. $|a - b| = |2 - 6| = |-4| = 4$ (or $|6 - 2| = 4$).

By conjunction introduction, since all components are true, the statement holds:

$$a > 0 \wedge b > 0 \wedge |a - b| = 4$$

- **Problem 2: Disjunction introduction.**

Consider the equation

$$x^3 - ax = 0,$$

where $a \in \mathbb{R}$.

Prove that, for every value of a , the equation has either exactly one or exactly three distinct real solutions. Determine for which values of a each case occurs.

- **Solution 2.1**

We can factor the given equation as:

$$x(x^2 - a) = 0$$

This gives one guaranteed real solution:

$$x = 0$$

The remaining solutions depend on the quadratic part:

$$x^2 = a$$

We separate this into cases based on the value of a :

- Case 1: If $a \leq 0$, then $x^2 = a$ has no distinct real solutions other than $x = 0$ (when $a = 0$, it is a repeated root). Thus, there is exactly one real solution.
- Case 2: If $a > 0$, then $x^2 = a$ yields two distinct real solutions: $x = \sqrt{a}$ and $x = -\sqrt{a}$. Since $a \neq 0$, these are distinct from $x = 0$. Thus, there are exactly three distinct real solutions.

Therefore, for any $a \in \mathbb{R}$, the equation has either exactly one solution (when $a \leq 0$) or exactly three solutions (when $a > 0$).

- **Problem 3: Disjunction elimination: proof by cases.**

Let $a_1, a_2, \dots, a_5 \in \mathbb{Z}$. Prove that among these five integers, one can always choose four numbers whose sum is even.

- **Solution 3.1**

Let K be the number of odd integers among the five given integers $a_1, a_2, \dots, a_5$.

The number of even integers is $5 - K$.

The possible values for K range from 0 to 5. We analyze the problem by partitioning K into the following three cases:

- **Case 1:** $K \leq 1$.

The number of even integers is $5 - K \geq 4$. We can choose 4 even numbers, which can be written as $2n_1, 2n_2, 2n_3, 2n_4$ for some $n_1, n_2, n_3, n_4 \in \mathbb{Z}$. Their sum is:

$$2n_1 + 2n_2 + 2n_3 + 2n_4 = 2(n_1 + n_2 + n_3 + n_4)$$

Since $n_1 + n_2 + n_3 + n_4 \in \mathbb{Z}$, the sum is even.

■ **Case 2:** $K \geq 4$.

There are at least 4 odd numbers available. We can choose 4 odd numbers, which can be written as $2n_1 + 1, 2n_2 + 1, 2n_3 + 1, 2n_4 + 1$ for some $n_1, n_2, n_3, n_4 \in \mathbb{Z}$. Their sum is:

$$(2n_1 + 1) + (2n_2 + 1) + (2n_3 + 1) + (2n_4 + 1) = 2n_1 + 2n_2 + 2n_3 + 2n_4 + 4 = 2(n_1 + n_2 + n_3 + n_4 + 2)$$

Since $n_1 + n_2 + n_3 + n_4 + 2 \in \mathbb{Z}$, the sum is even.

■ **Case 3:** $2 \leq K \leq 3$.

There are at least 2 odd numbers and at least 2 even numbers available. We can choose 2 even numbers ($2n_1, 2n_2$) and 2 odd numbers ($2n_3 + 1, 2n_4 + 1$) for some $n_1, n_2, n_3, n_4 \in \mathbb{Z}$. Their sum is:

$$2n_1 + 2n_2 + (2n_3 + 1) + (2n_4 + 1) = 2n_1 + 2n_2 + 2n_3 + 2n_4 + 2 = 2(n_1 + n_2 + n_3 + n_4 + 1)$$

Since $n_1 + n_2 + n_3 + n_4 + 1 \in \mathbb{Z}$, the sum is even.

Since all possible values of K fall into one of these three cases, and each case mathematically guarantees an even sum, the proof is complete.

• **Problem 4**

Construct a truth table and prove that

$$(P \rightarrow Q) \wedge (P \rightarrow R) \equiv P \rightarrow (Q \wedge R).$$

◦ **Solution 4.1**

The truth table for both expressions is constructed below:

P	Q	R	$P \rightarrow Q$	$P \rightarrow R$	$(P \rightarrow Q) \wedge (P \rightarrow R)$	$Q \wedge R$	$P \rightarrow (Q \wedge R)$
1	1	1	1	1	1	1	1
1	1	0	1	0	0	0	0
1	0	1	0	1	0	0	0
1	0	0	0	0	0	0	0
0	1	1	1	1	1	1	1
0	1	0	1	1	1	0	1
0	0	1	1	1	1	0	1
0	0	0	1	1	1	0	1

Since the columns for $(P \rightarrow Q) \wedge (P \rightarrow R)$ and $P \rightarrow (Q \wedge R)$ are identical in every row, the two statements are logically equivalent.

• **Problem 5**

Let the universe of discourse be

$$X = \{a, b\},$$

and let $P(x)$ and $Q(x)$ be mathematical statements depending on $x \in X$.

Using a truth table, determine whether the statements

$$\forall x \in X (P(x) \wedge Q(x))$$

and

$$(\forall x \in X P(x)) \wedge (\forall x \in X Q(x))$$

are logically equivalent.

First rewrite each quantified statement using only $P(a)$, $P(b)$, $Q(a)$, and $Q(b)$, and then construct the truth table.

◦ **Solution 5.1**

First, we expand the quantified statements over the finite domain $X = \{a, b\}$:

$$1. \quad \forall x \in X (P(x) \wedge Q(x)) \equiv (P(a) \wedge Q(a)) \wedge (P(b) \wedge Q(b))$$

$$2. \quad (\forall x \in X P(x)) \wedge (\forall x \in X Q(x)) \equiv (P(a) \wedge P(b)) \wedge (Q(a) \wedge Q(b))$$

By the associative and commutative laws of conjunction ($\wedge$), both expressions simplify directly to:

$$P(a) \wedge P(b) \wedge Q(a) \wedge Q(b)$$

To keep the table compact and readable, let

$$\begin{aligned} A &\equiv P(a) \wedge Q(a) & B &\equiv P(b) \wedge Q(b) & C &\equiv \forall x \in X (P(x) \wedge Q(x)) \\ D &\equiv \forall x \in X P(x) & E &\equiv \forall x \in X Q(x) & F &\equiv (\forall x \in X P(x)) \wedge (\forall x \in X Q(x)) \end{aligned}$$

Thus, they are logically equivalent. The corresponding truth table configuration for their components is:

$P(a)$	$P(b)$	$Q(a)$	$Q(b)$	A	B	C	D	E	F
1	1	1	1	1	1	1	1	1	1
1	1	1	0	1	0	0	1	0	0
1	1	0	1	0	1	0	1	0	0
1	1	0	0	0	0	0	1	0	0
1	0	1	1	1	0	0	0	1	0
1	0	1	0	1	0	0	0	0	0
1	0	0	1	0	0	0	0	0	0
1	0	0	0	0	0	0	0	0	0
0	1	1	1	0	1	0	0	1	0
0	1	1	0	0	0	0	0	0	0
0	1	0	1	0	1	0	0	0	0
0	1	0	0	0	0	0	0	0	0
0	0	1	1	0	0	0	0	1	0
0	0	1	0	0	0	0	0	0	0
0	0	0	1	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0

Therefore, the statements are logically equivalent.